

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals and Data Structures	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated: 21st May, 2026
Semester: 1st	Session: 2026
Lab/Project/Assignment #: 15	CLOs to be covered: CLO3
Lab Title: GUI using Qt Designer	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO3	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Mark s	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 15: GUI using Qt Designer

Lab Goals/Objectives:

To design and implement a graphical user interface (GUI) using PyQt5, including labels, input fields, and buttons, and to connect the interface to a running application.

Equipment Required:

Python 3, PyQt5, Qt Designer, VS Code / PyCharm

Theory:

PyQt5 is a set of Python bindings for the Qt application framework, used to build graphical user interfaces. Widgets such as QLabel, QLineEdit and QPushButton can be arranged using layouts (e.g. QVBoxLayout) inside a QWidget or QFrame to build interactive forms.

Lab Work:

Task 1: PyQt5 Registration/Login Form

```
from PyQt5 import QtCore, QtGui, QtWidgets
class Ui_Form(object):
    def setupUi(self, Form):
        Form.setObjectName("Form")
        Form.resize(488, 640)
        Form.setStyleSheet("background-color: rgb(145, 190, 210);")
        # University logo label
        self.label_6 = QtWidgets.QLabel(Form)
        self.label_6.setGeometry(QtCore.QRect(150, 10, 180, 140))
        self.label_6.setMinimumSize(QtCore.QSize(130, 130))
        self.label_6.setMaximumSize(QtCore.QSize(250, 250))
        self.label_6.setText("")

self.label_6.setPixmap(QtGui.QPixmap("../Pictures/Screenshots/University_of_Engineering_and_Technology_Lahore_logo.svg.png"))
self.label_6.setScaledContents(True)
self.label_6.setObjectName("label_6")
# Empty label (placeholder)
self.label_5 = QtWidgets.QLabel(Form)
self.label_5.setGeometry(QtCore.QRect(140, 10, 111, 51))
self.label_5.setText("")
self.label_5.setObjectName("label_5")
# University name label
self.label = QtWidgets.QLabel(Form)
self.label.setGeometry(QtCore.QRect(10, 140, 471, 51))
self.label.setStyleSheet('font: 57 italic 23pt "Z003";')
self.label.setObjectName("label")
# White card frame
self.frame = QtWidgets.QFrame(Form)
self.frame.setGeometry(QtCore.QRect(90, 220, 303, 281))
self.frame.setStyleSheet("background-color: white;\n border-radius: 10px")
self.frame.setFrameShape(QtWidgets.QFrame.StyledPanel)
self.frame.setFrameShadow(QtWidgets.QFrame.Raised)
self.frame.setObjectName("frame")
# Vertical layout inside frame
self.verticalLayout = QtWidgets.QVBoxLayout(self.frame)
```

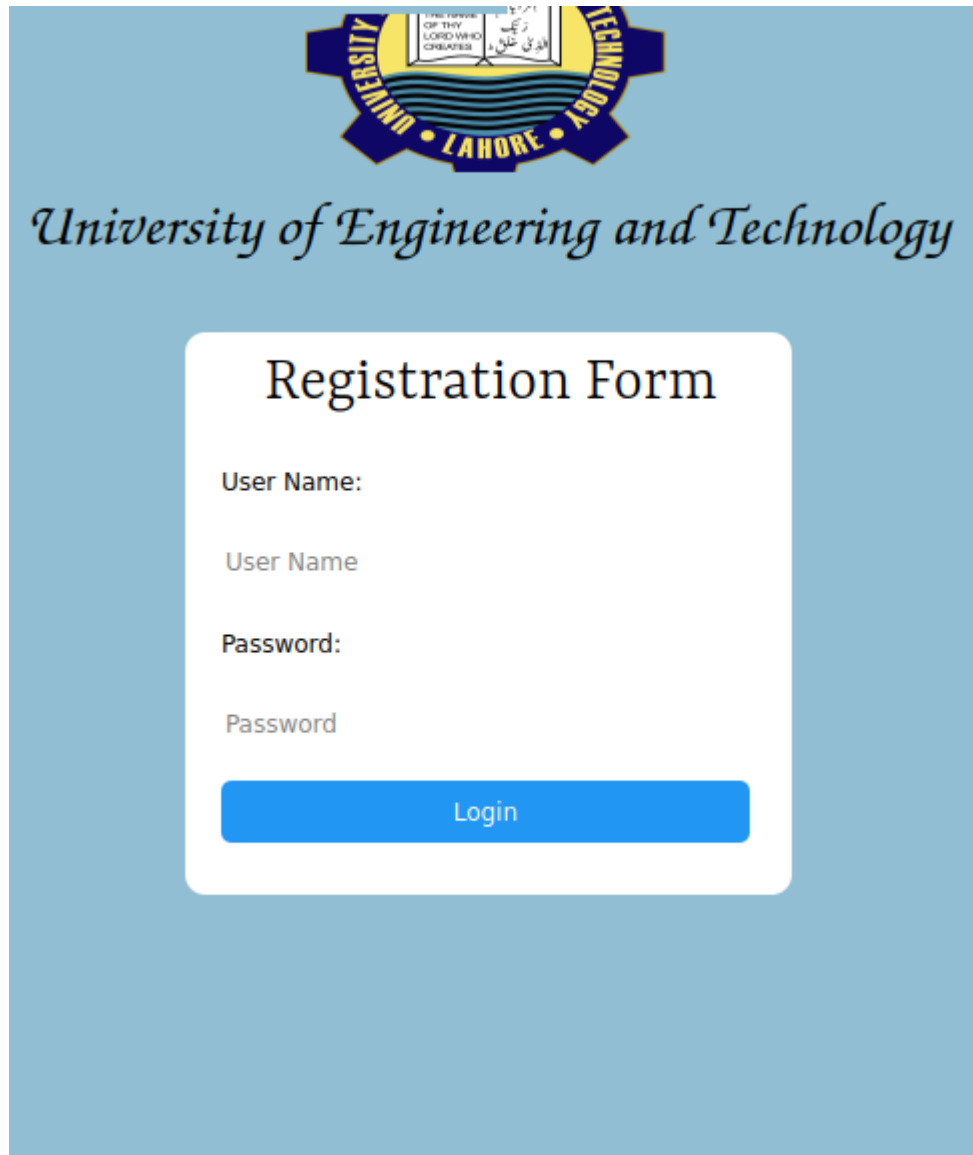
```

self.verticalLayout.setSpacing(20)
self.verticalLayout.setContentsMargins(18, 9, 21, 26)
self.verticalLayout.setObjectName("verticalLayout")
# Registration Form title
self.label_2 = QtWidgets.QLabel(self.frame)
self.label_2.setStyleSheet('font: 75 24pt "Yrsa";')
self.label_2.setObjectName("label_2")
self.verticalLayout.addWidget(self.label_2)
# User Name label
self.label_3 = QtWidgets.QLabel(self.frame)
self.label_3.setObjectName("label_3")
self.verticalLayout.addWidget(self.label_3)
# User Name input
self.lineEdit = QtWidgets.QLineEdit(self.frame)
self.lineEdit.setPlaceholderText("User Name")
self.lineEdit.setObjectName("lineEdit")
self.verticalLayout.addWidget(self.lineEdit)
# Password label
self.label_4 = QtWidgets.QLabel(self.frame)
self.label_4.setObjectName("label_4")
self.verticalLayout.addWidget(self.label_4)
# Password input
self.lineEdit_2 = QtWidgets.QLineEdit(self.frame)
self.lineEdit_2.setEchoMode(QtWidgets.QLineEdit.Password)
self.lineEdit_2.setPlaceholderText("Password")
self.lineEdit_2.setObjectName("lineEdit_2")
self.verticalLayout.addWidget(self.lineEdit_2)
# Login button
self.pushButton = QtWidgets.QPushButton(self.frame)
self.pushButton.setStyleSheet(
    "background-color: #2196F3;\n"
    "color: white;\n"
    "border-radius: 5px;\n"
    "padding: 8px;")
self.pushButton.setObjectName("pushButton")
self.verticalLayout.addWidget(self.pushButton)
self.retranslateUi(Form)
QtCore.QMetaObject.connectSlotsByName(Form)
def retranslateUi(self, Form):
    _translate = QtCore.QCoreApplication.translate
    Form.setWindowTitle(_translate("Form", "Form"))
    self.label.setText(_translate("Form", "University of Engineering and Technology"))
    self.label_2.setText(_translate("Form", " Registration Form"))
    self.label_3.setText(_translate("Form", "User Name:"))
    self.label_4.setText(_translate("Form", "Password:"))
    self.pushButton.setText(_translate("Form", "Login"))
if __name__ == "__main__":
    import sys
    app = QtWidgets.QApplication(sys.argv)

```

```
Form = QtWidgets.QWidget()
ui = Ui_Form()
ui.setupUi(Form)
Form.show()
sys.exit(app.exec_())
```

Result:



The image shows a screenshot of a registration/login form for the University of Engineering and Technology (UET) Lahore. The form is titled "Registration Form" and is set against a light blue background. At the top, there is a logo of the university, which features a shield with a book and the text "UNIVERSITY OF ENGINEERING AND TECHNOLOGY LAHORE". Below the logo, the university's name is written in a stylized font. The form itself is a white rounded rectangle containing the following elements:

- A label "User Name:" followed by an input field.
- A label "Password:" followed by an input field.
- A blue button labeled "Login" at the bottom.

Conclusions:

A PyQt5-based registration/login form was implemented using labels, input fields, a styled frame, and a button, demonstrating the fundamentals of GUI design with Qt.